

Atom Library — Common Functions

Function	Domain	Curvature	Monotonicity	Range
$ x $	$\mathbb{R}$	convex	nonmonotone	$[0, \infty)$
x^2	$\mathbb{R}$	convex	nonmonotone	$[0, \infty)$
$\max(x_1, x_2)$	$\mathbb{R}^2$	convex	nondecreasing	$\mathbb{R}$
$\exp(x)$	$\mathbb{R}$	convex	nondecreasing	$(0, \infty)$
$\log(x)$	$(0, \infty)$	concave	nondecreasing	$\mathbb{R}$
$\sqrt{x}$	$[0, \infty)$	concave	nondecreasing	$[0, \infty)$
$\min(x_1, x_2)$	$\mathbb{R}^2$	concave	nondecreasing	$\mathbb{R}$
$\ \mathbf{x}\ _p$	$\mathbb{R}^n$	convex	nonmonotone	$[0, \infty)$